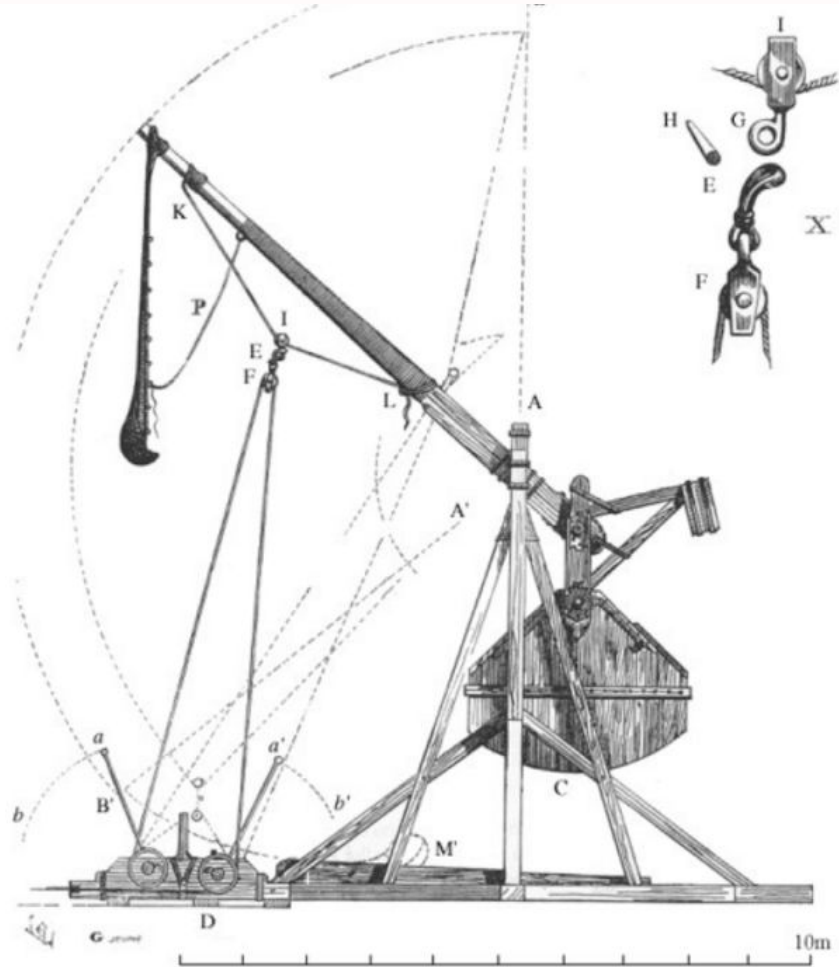




Designing A Counter Weight Trebuchet

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Table of Contents

Project Definition and Aim	3
Background Information:	3
Project Definition and Aim:	3
Design Theory and Methodology	4
Design Mechanism and Calculations	6
Mechanism Description of The Trebuchet:	6
Discussion & Calculation:	6
Device Description	8
Bill of Materials:	8
BASE 1 & BASE 2	9
Base 1:	9
Base 2:	9
More Details:	10
LOW-SIDE PART & SIDE PART	10
Low-side Part:	10
Side Part:	11
More Details:	11
FULCRUM	11
POUCH	12
FINGER	12
LEVEL ARM	13
Long Arm:	13
Short Arm:	13
Assembling:	14
References	15
Appendices	16

I. Project Definition and Aim

Background Information:

Historical paper indicated that the Southern cities were surrounded by mountains on 3 sides and Han river on 1 side, that made the spot difficult for Mogol to take. Therefore, the weapon with the ability to shoot across long distances is the key weapon for Mongol to win the battle with the Song Empire. In this time, Mogol had hired Persian engineers to produce some trebuchets which could launch the projectile *50kg* in the range of *100 meters*. They brought a terrible destruction to the Song's defense castle system. As a result, Kublai Khan who led Mongol emperors founded the Yuan Dynasty, whilst their opponents in the Southern Song Empire completely collapsed.

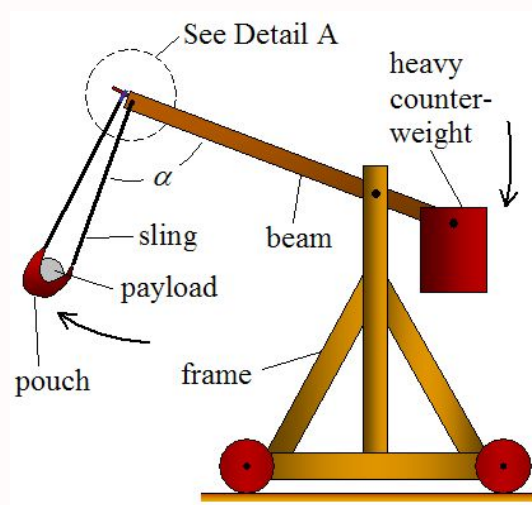


Figure 1.1: A trebuchet model

Project Definition and Aim:

Our team with 6 members, who assumed to be time travelers to Xiangyang in AD 1267, which is Hubei province as nowadays. Our mission is building those instead the mentioned engineer team. We consulted Kublai Khan to manufacture the weapons within 4 weeks, that were requested with the specific constraints:

- The weapon is able to launch the spherical projectile with a diameter of 45 mm and weighs 2.7 grams with a minimum distance of 5 meters.

- It should hang at least 2 PASCO hooked masses with a total 1 kilogram as counterweight.
- Trebuchet should be able to adjust for shooting at different ranges and angles to destroy the target.
- It must not have more than 12 parts, with each part fit into a 14x14x14 centimeters box.
- It includes maximum 4 parts made from Tough Resin material and the remaining manufactured by Acrylic material.

During the time we designed and built the project, we found some significant challenges which slow down our progress. In detail, the terminal velocity which the projectiles leave out the trebuchet is hard to find out. Moreover, it hardly calculated and estimated the length of sling to fire the furthest distance. Last but not least, there is a difficulty to control the angle of firing that different angles would lead to the different distances which directly impact our final result.

This report is written as a technical document, which shows step-by-step how the device is designed and which components it accommodates. The main objective of the project is mentioned as above. Therefore, there are parts that need to hold the 19.6N going downward. The project helps us to be ready for the engineering project in the industry. This project allows us to understand the norm, the boundaries of engineering practices, and utilize tools and resources and techniques to solve problems.

This section explains the requirements and engineering specifications were initially generated by our members and weighted for importance by using the virtual trebuchet. In order to fire the ping pong ball over 5 meters, we decided to make a hinge trebuchet (*counterweight is hung on the short arm*) with a sling at the end of the long arm. Improvements and tweaking to fit in realism after presenting the first prototype trebuchet's parts, assembly, and theoretical calculations to our lecturer.

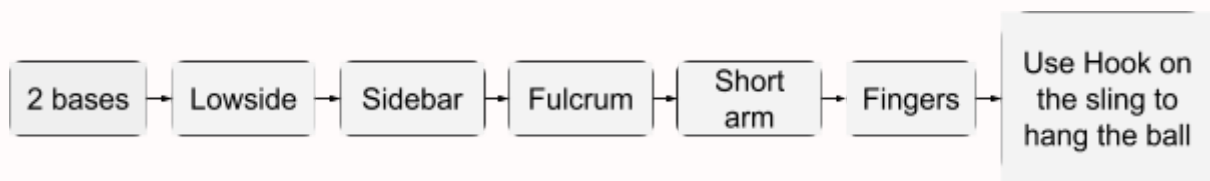


Figure 1.2: Designing a trebuchet step-by-step

II. Design Theory and Methodology

The counter weight trebuchet launching projectile base on the conservation energy theory. On the design physics analysis, it is assumed that the pouch, sling weight can be ignored and there is no wind resistance affecting the ping pong ball.

$$KE_f + PE_f = KE_i + PE_i \quad [6]$$

The conservation of energy theory states that energy from the weight fall is transferred through the level arm to the projectile at the launching end of the arm in a close system. In the counter weight trebuchet design, the weight is transferred from kinetic rotational energy:

$$K_r = \frac{1}{2} I \omega^2$$

[5]

When the projectile is dragged in the base to the launching point, the object has drag efficiency. The ping pong ball energy is increased due to its sphere shape and increases the maximum launching distance. In order to maximize using it, the projectile sling is recommended to be at 80% of the long arm [5]. From all the theory above, it can be illustrated the counter weight trebuchet design theory. The weight is released at a certain high from the base, it will accelerate as it is falling and the level arm rotation will transfer energy to the projectile end. On the projectile end, the ping pong ball energy will increase as it sling gets longer. The ping pong ball shooting angle is dependent on the finger at the top of the level arm.

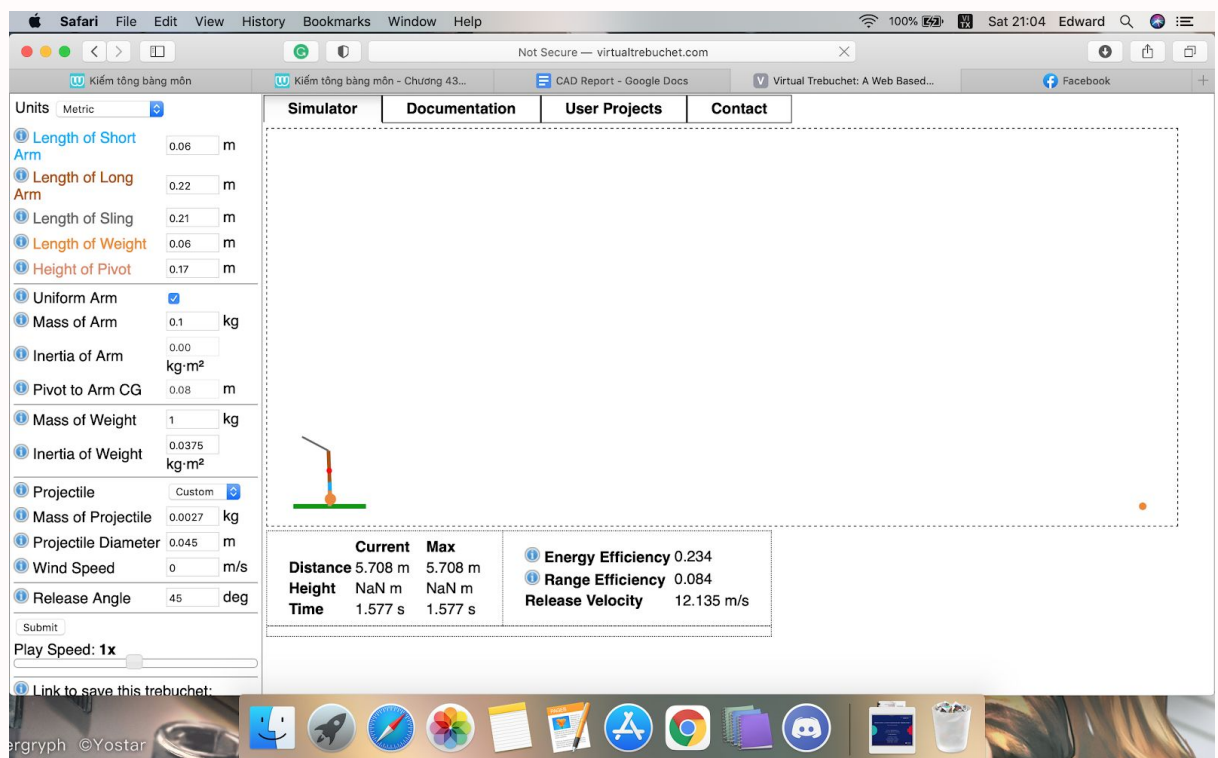


Figure 2.1 Trebuchet simulation

At first, the height of pivot is decided, after calculation and estimate that weight rotational energy is fully converted without the pasco hook mass hitting the base, pivot height is selected at 17cm. After that, the level arm is calculated using pytago theory as the base is selected to have 14cm in length. The level long arm is 22cm and short arm is 5.8 cm with the following 3.75:1 ratio [5], through test and trial, it is discovered that the short arm with 6cm in length will shoot at further range so the level arm dimension is decided. The control the projectile firing range a finger at the top of the level arm is designed, the finger is bench from

90-180 degree to allow the launching range to reach 45 degree at the most optimized condition. The sling length on both projectile and weight is contributed in the shooting range, long arm sling is decided to be 21cm long and weight sling is 6cm to avoid pasco hook mass hitting the base and losing energy.

III. Design Mechanism and Calculations

Mechanism Description of The Trebuchet:

The trebuchet in this project is a device that is used to throw a spherical projectile upto a distance. The trebuchet transforms potential energy of falling weight into kinetic energy inside the projectile to launch it. This trebuchet follows the design requirements to be able to launch a spherical projectile with diameter up to 45 mm and weight 2.7 grams over a distance of at least 5 meters.

The trebuchet in this design is composed of 2 parallel vertical plates (see Figure 4.3) on 2 bases (see Figure 4.1- 4.2), on top of these 2 plates are 2 rectangle plates (see Figure 4.4) attached to them. There is a cylinder bar connecting the 2 rectangle plates at their apices (see Figure 4.5), which works as an axis of rotation for the arm. The arm is assembled from 2 rectangle plates, namely long arm and short arm (see Figure 4.8- 4.9), there is an L rod extending from one end of the long arm (see Figure 4.10).

Discussion & Calculation:

The level arm

The level arm is the part that transforms potential energy of the heavy weight, which would be hanged at the end of the short arm, into rotational kinetic energy of the projectile to launch it.

The arm is separated into the short arm and the long arm. The short arm has one end consisting of a hole for hanging the heavy weight, which stores potential energy for the device to work. The long arm has an L rod extending at one end to hang the sling, which also has a pouch attached, to contain and throw the projectile (see Figure 1.1). The length of the level arm is 28cm, which is the combine of the 6cm short arm and the 22cm long arm. The ratio of the length of the long arm over the short arm is 3.67:1, which is close to 3.75:1 to maximize the efficiency of the trebuchet [5]. The level arm is 5mm thick, enough to withstand more than 3kg weight and also fit the design requirements.

The finger

The finger is an L-shaped steel rod with a 2cm edge and a 3cm edge. The angle between the 2 edges should be 51 degree in order to launch the projectile at around 45 degree from horizontal to maximize efficiency [5].

The calculation for the angle between the 2 edges is as follow:

- μ : coefficient of friction

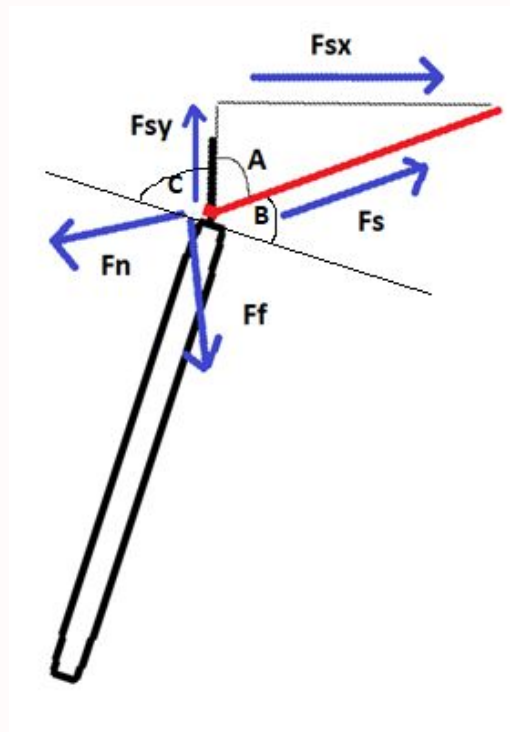


Figure 3.1: Free-body diagram of finger and sling

$$F_n = F_{sx} = F_s \cdot \sin A$$

$$F_f = \mu \cdot F_n = \mu \cdot F_{sx} = \mu \cdot F_s \cdot \sin A$$

$$F_{sy} = F_s \cdot \cos A$$

At release: $F_{sy} = F_f$

$$\Leftrightarrow F_s \cdot \cos A = \mu \cdot F_s \cdot \sin A$$

$$\Leftrightarrow A = \arctan(1/\mu)$$

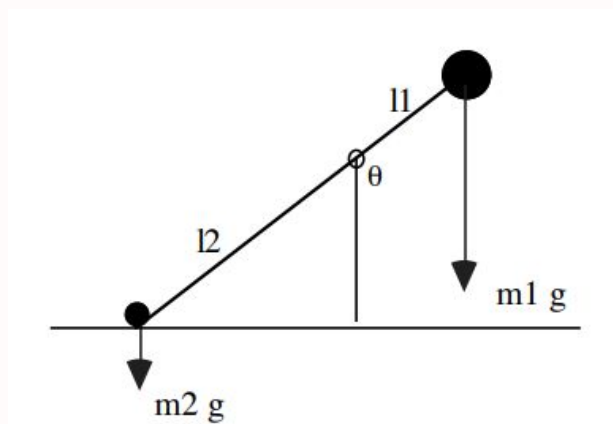
The coefficient of friction of nylon string, which is the material of the sling, equal to 0.1 [6]

$$\Rightarrow A = 84 \text{ degree}$$

If $B = 45 \text{ degree}$ (launch at 45 degree)

$$\Rightarrow C = 180 - 45 - 84 = 51 \text{ degree}$$

Calculation



$$\tau = I\omega = l * F = m1 * l1 * g * \sin\theta - m2 * l2 * g * \sin\theta$$

$$I = m1 * l1^2 + m2 * l2^2$$

$$\alpha = \frac{\tau}{I}$$

$$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$$

$$v = l * \omega$$

Put in all the dimensions of our trebuchet, we have the result of $v = 5,249$ m/s, which has high error, up to 44%. But this is due to the energy efficiency of practice when compared to theory. It could be sling or other factor that have not been included in the formula affecting the results.

IV. Device Description

Bill of Materials:

Part No	Description	Quantity	Materials	Method of compliance (3D print, laser cut)
1	Base 1	1	Tough Resin	3D print
2	Base 2	1	Tough Resin	3D print
3	Long arm	1	Tough Resin	3D print
4	Short arm	1	Tough Resin	3D print
5	Low-side part	2	Acrylic	Laser cut
6	Side part	2	Acrylic	Laser cut
7	Finger	1	Steel	From market (team self supply)
8	Fulcrum	1	Steel/Carbon fiber	From market (team self supply)

BASE 1 & BASE 2

Base 1:

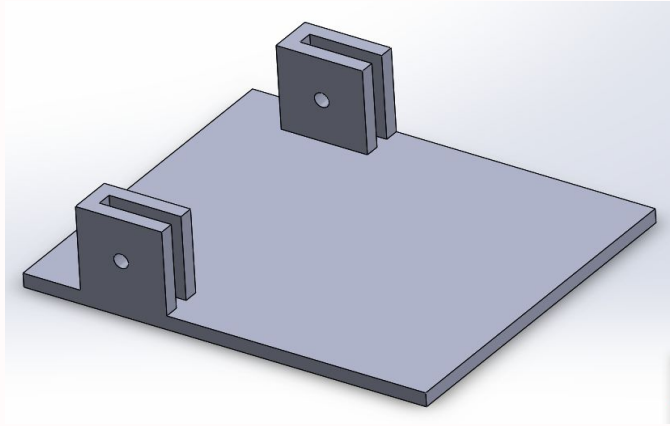


Figure 4.1: Base 1 3D picture

It is a rectangle piece of tough material that is 14cm long and 12cm wide with a thickness of 5mm. This base is decided to be manufactured by a 3D printing process. Two square joints which have 5mm thickness are located 2cm from the left edge. A space inside the joints is expected to be 5mm wide plus with a 0.1mm allowance so as to fit a 5mm thick low-side part. There is a through-hole with a diameter of 0.51mm in the center of each joint. They are created for screwing to connect the base with the low-side part tightly.

Base 2:

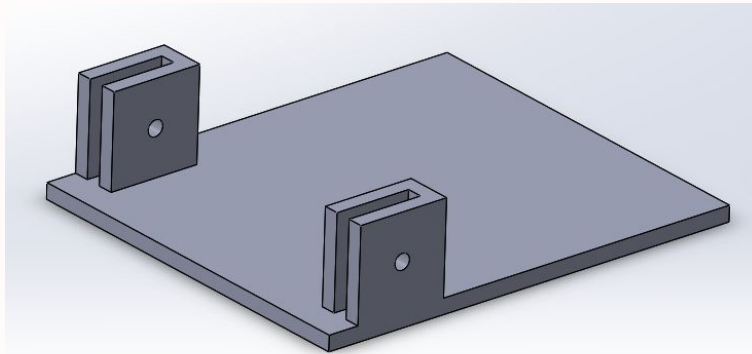


Figure 4.2: Base 2 3D picture

It is another rectangle piece of tough material with similar dimensions to Base 1. It is also made by a 3D printing process. There are two opposite joints that also have the same features as the ones on Base 1. The distance from joints on Base 1 to the right edge of Base 1 is 11.5cm, so two joints in this base should be located 3.5cm from the left edge. Thus, those joints will fit a 15cm long low-side part which is demonstrated below. The blank space on the right has special potential. It is not only the surface for the long arm to touch, but it also keeps the trebuchet more steady and prevents it from overturning when it launches the projectile.

More Details:

Each base has a volume of 102ml, 14cm length and 12cm width. So that the bases meet the teacher's requirement. (less than 150ml ; fit in a 14cmx14cmx14cm box). All holes have an allowance of 0.1mm in order to fit with the other part easily. Since the joints on the bases have some complicated features, the bases must be 3D printed instead of laser cutting. (laser cut only cut on the front face). We use maximum length for each base. As a result, we can make a 24cm long base. So, in general, the other parts could be made in a bigger size, which leads to a bigger trebuchet. It may help the trebuchet launch the projectile further easily. In this case, the width of each base is 12cm. After subtracting the widths of two joints, it remains about 9cm width of empty space. Thus, there is enough space for even 2 PASCO 500g Hooked Mass to go through (each one is 43cm wide).

LOW-SIDE PART & SIDE PART

Low-side Part:

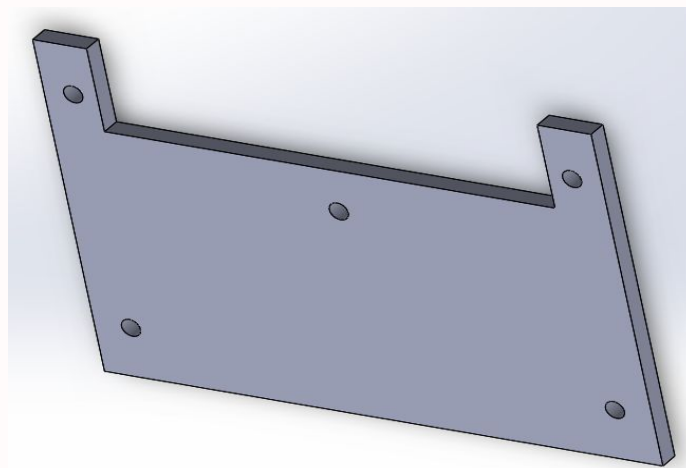


Figure 4.3: Lowside 3D picture

It is a rectangle piece of Acrylic, which is 5mm thick, 15cm long and 10cm wide. It is decided to be made by using a Laser Cutter. There is a rectangle part through the cut, which is 12.1cm long and 2.5cm wide. " 0.1mm long " is the allowance to fit the joint easily. The empty space is to join the side part to the low-side one. There are two-side holes and the middle one which are placed around the empty space. They help to fix two parts together by some kinds of fasteners. There are also two holes on two sides at the bottom for screwing, which are mentioned before.

Side Part:

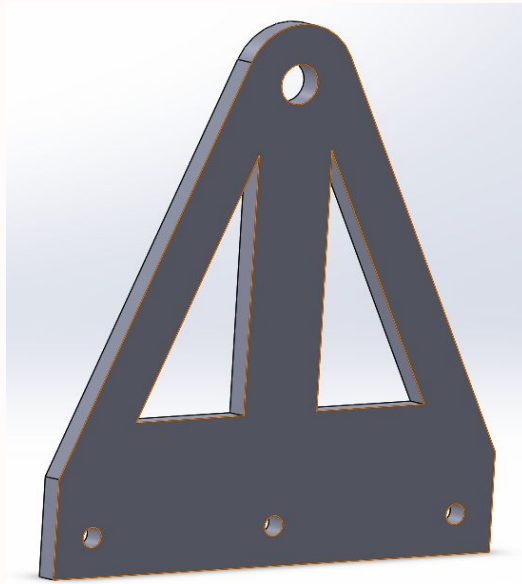


Figure 4.4: Side part 3D picture

It is an isosceles triangle piece with two right triangle holes on two sides adjoining a rectangle part which is 12cm long and 2.5cm wide. This part is also decided to be a Laser Cut. The rectangle part will be fixed in the space of the low-side part. There is a 1.01 diameter hole on the top to put the fulcrum through. The right triangle holes are cut to reduce the amount of solid in part; they also create three bars instead. As a result, when the trebuchet launches a projectile, the force will be applied to both three bars instead of the fully solid isosceles triangle piece. That can reduce the Inertia created. In addition, reducing the solid means the side part's weight will decrease, which leads to the reduction of Inertia while launching. The less Inertia created, the less risk of the trebuchet overturning.

More Details:

The distance from the 1.01 diameter hole to the bottom edge of the isosceles triangle part is 9.5 cm, plus with 7.5 cm remaining of the low-side part, so the total length of pivot will be 17cm. It is the same dimension as what we calculated. Because the low-side parts and side parts only have details that can be cut on the front face, they are put into the laser-cut list.

FULCRUM

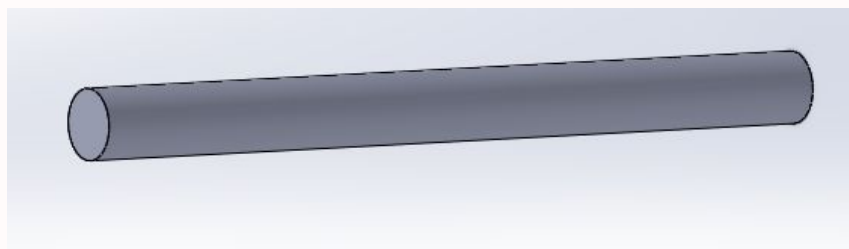


Figure 4.5: Fulcrum 3D picture

The Fulcrum is a small diameter bar which provides the axis for the trebuchet lever arm to spin around it. The fulcrum is constructed to be a cylinder shape supporting the level arm spinning on it freely with the less friction caused from two surfaces, the level hole and fulcrum surface. Its diameter is 10mm and the bar length is 17cm, the bar 10mm diameter will support the level arm projectile and weight from breaking the fulcrum or affecting the level arm movement, the two spare spaces on both sides of fulcrum are contributing to it stationary with rubber rings. The fulcrum bar is made of steel and it can be bought from the market.

POUCH

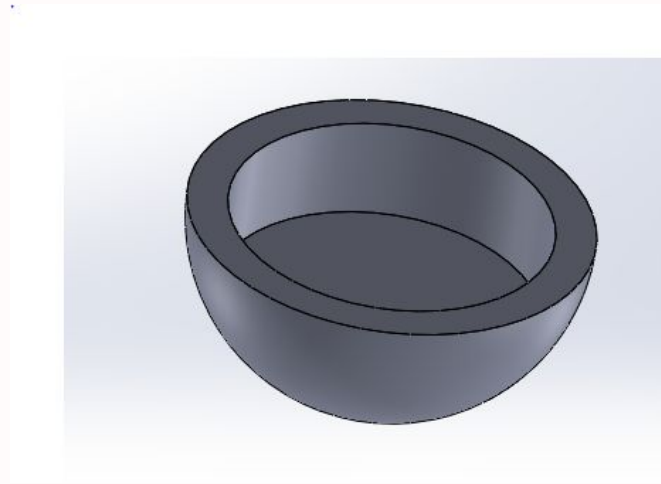


Figure 4.6: Pouch 3D picture

The figure above is an illustration of a ping pong ball holder, the pouch material varies from cloth to nylon. It can be constructed with either hard or soft materials. Its primary objective is to hold ping pong balls until it reaches designated launching point, it is designed to be lightweight in order not to affect projectile velocity and movement. Its diameter is greater than 45mm to have enough space holding the ping pong ball and attach string connecting to the trebuchet lever arm.

FINGER

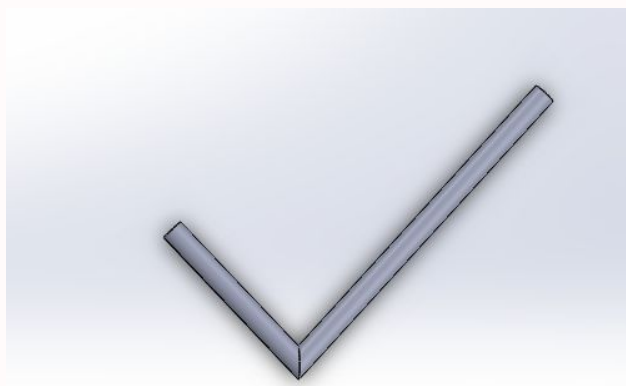


Figure 4.7: Finger 3D picture

The finger is the most important component within the trebuchet to be able to control the machine firing angle. The finger is constructed with two parts, the lower half to fit in the hole at the top of the level arm and the upper part to control shooting angle. The whole finger is in the shape of a cylinder which can be bought from the market with its material is made of steel or aluminum. The lower half is 2cm long in order to be stationary inside the hole. The upper part is benched around 90-180 degrees to the lower half and 3cm long, through simulation and calculation the finger degrees lower than 90 will launch projectiles at a negative angle and when finger create an shooting angle is 45 degree the trebuchet machine will launch the projectile at its max distance is 5.2m. On the second part of testing and building the trebuchet, the finger can be justified to the desired angle by using tools to change the finger upper half degrees.

LEVEL ARM

Long Arm:

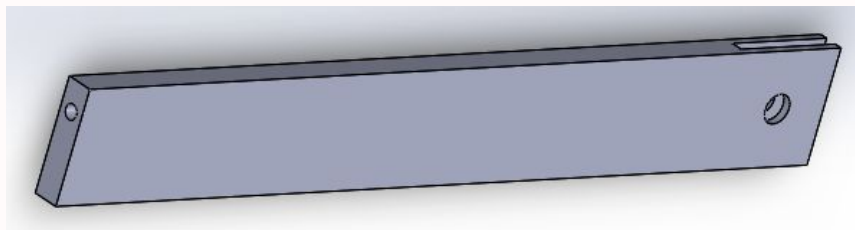


Figure 4.8: Long arm 3D picture

The long arm design is constructed using 3D printing tough resin material. It must fit in a 14x14x14 box and the volume cannot exceed 150ml. The long arm has 16cm length and 5mm thickness, it can be printed using a 3D printer. The long arm has a 2mm hole on its top face which allows putting in the finger components, and a 0.51 diameter hole to connect the short arm. The long arm components use 5mm thickness that can withstand more than 3kg weight, and will be stationary after shooting projectiles. On the bottom of the long-arm, a small square 3mm box and 2cm depth is cut fit to create an hole that can join the short-arm and long-arm together, the small hole is for screws stationary two parts to avoid the arm will drop out after launching projectiles.

Short Arm:

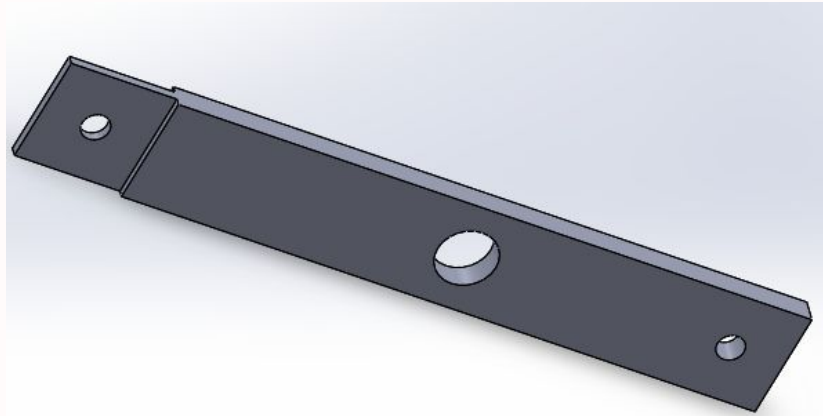


Figure 4.9: Short arm 3D picture

The figure demonstrated above is level arm short-arm, it is including two parts: the connection square box and the left part of the level arm. The connection square box has 3mm thickness and 2 cm long, it has an 0.51 diameter hole at the center to join with a long arm. The 0.51 diameter hole at the short arm bottom is used for putting rope holding weight objects. On the left part of the level arm, the 1.01 diameter hole is fitting in for fulcrum to allow the whole level arm spinning around it. After many trials and testing the fulcrum hole at 6cm to the bottom divided the level arm to projectile arm has 22cm length and weight arm has 6cm length shooting the most distance base on the current constructing counter-weight trebuchet. The short-arm is manufactured using a 3D print machine and its volume is 14.2 ml.

Assembling:

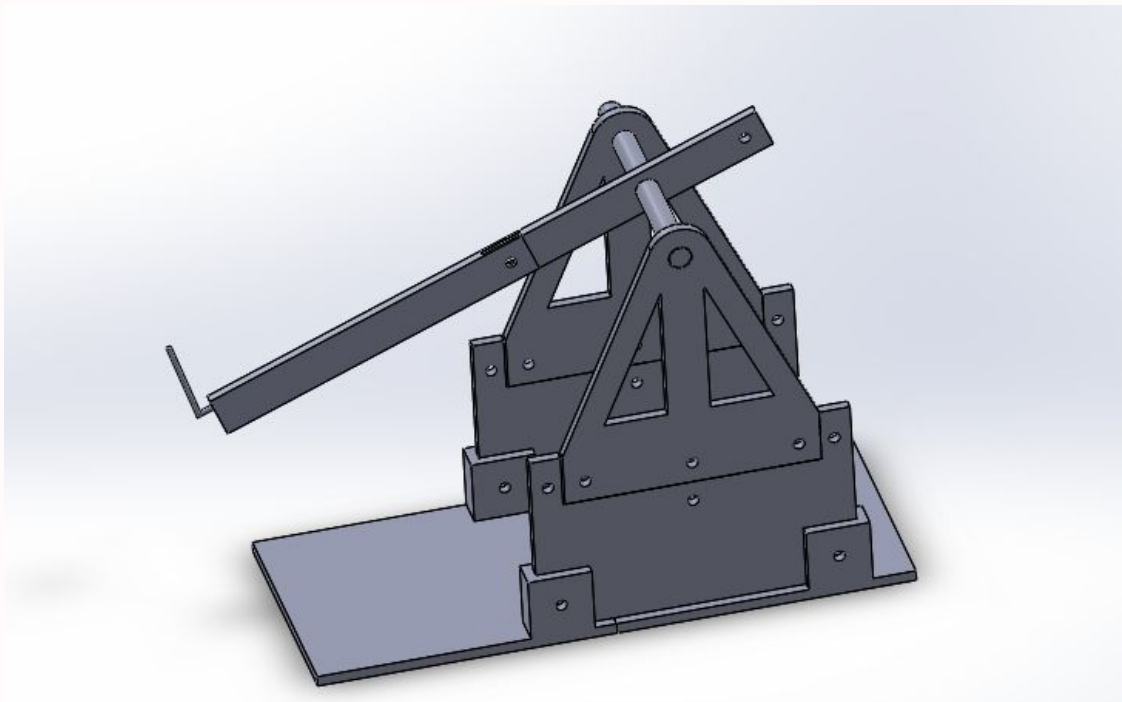


Figure 4.10: Trebuchet assembling

After collecting all parts in the design and getting a string to hold the projectile and weight. The trebuchet will be assembled and tested at the maximum launching range. First, the base 1 and base 2 will connect through the two lowside parts, in order to join three components together a 5mm screws are used. After that, the side part will join with the lowside, the two parts hole will use a cropper wire or a small plastic wire to stationary them. The long arm and short arm are connected with the join part between it and holding it together and concentrated with the fulcrum. The fulcrum is put in the top hole of the side part and is attached with the rubber rings at both sides to avoid it moving and the level arm at the center of the fulcrum will also be stationary with rubber rings. The last part is finger, it is attached to the top of the level arm using two faces tape. The finger degree can be adjusted using tools to the desired shooting angle, after that, the finger will be used with super glue attached to the top level arm.

V. References

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VI. Appendices

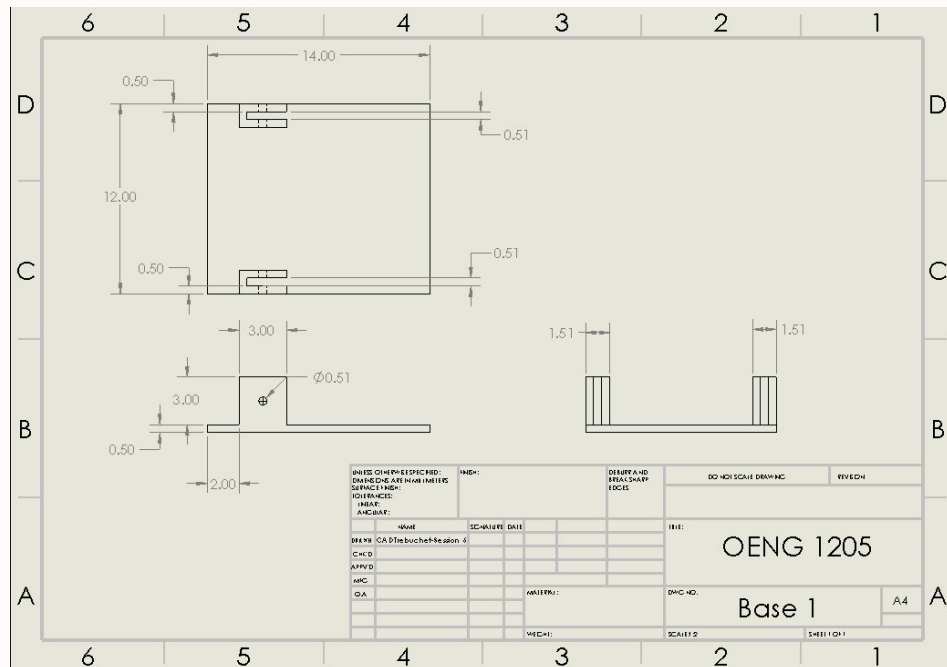


Figure 6.1: Base 1 drawing

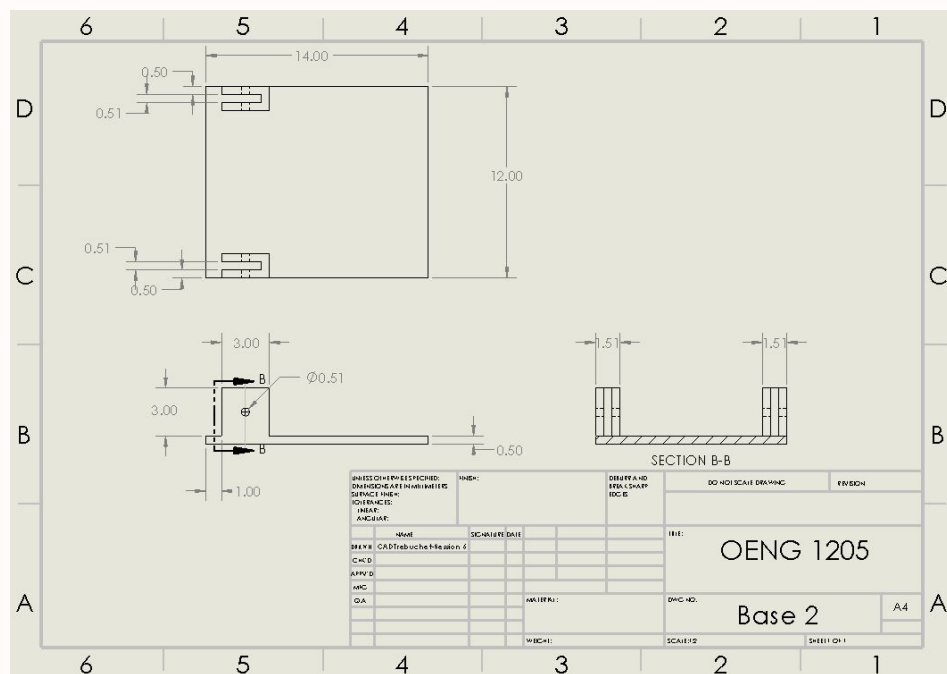


Figure 6.2: Base 2 drawing

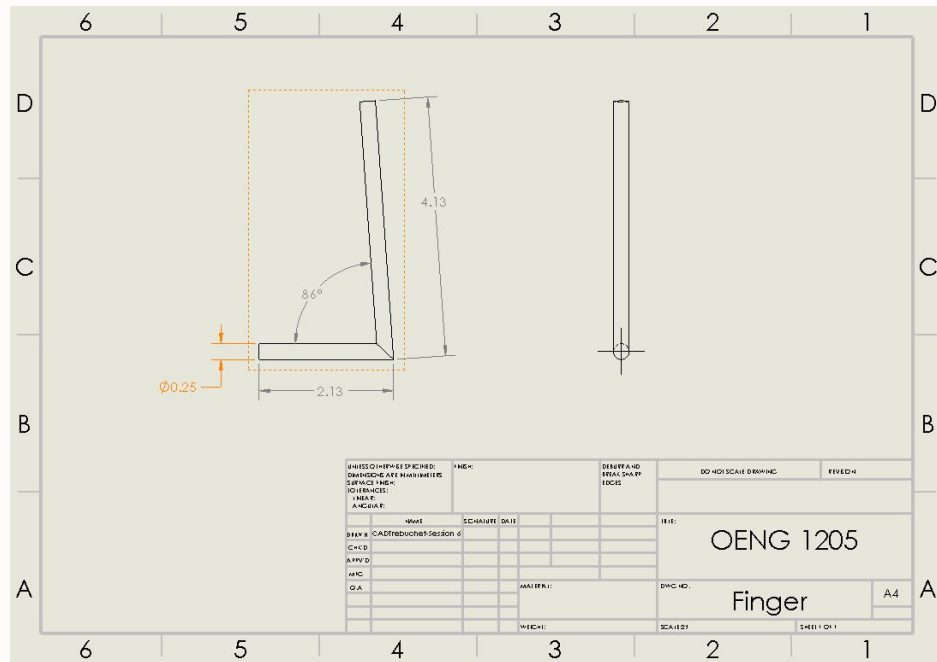


Figure 6.3: Finger drawing

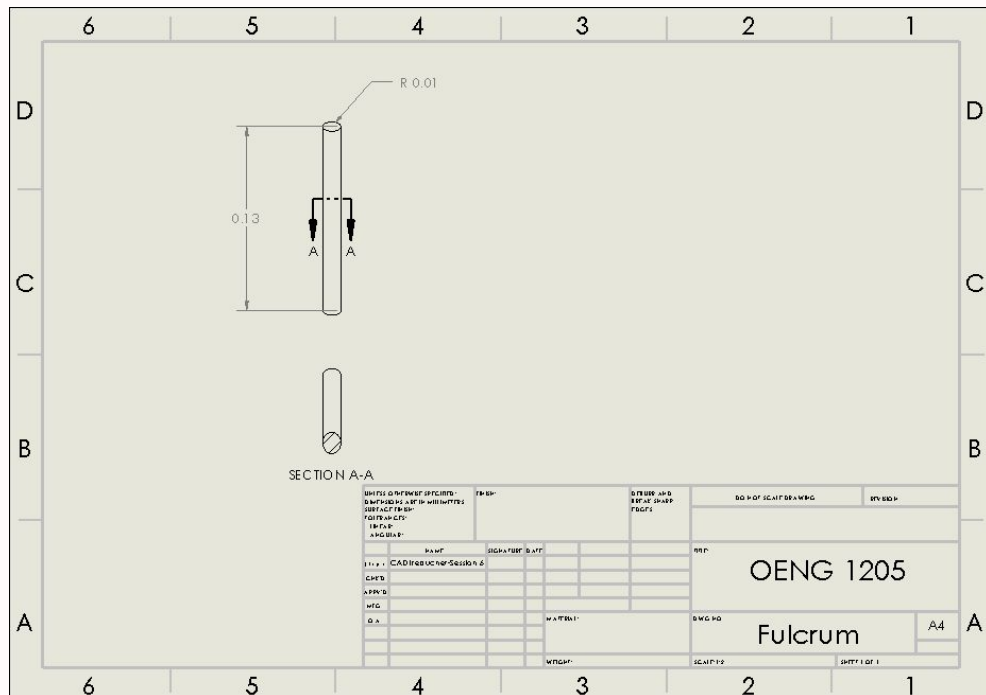


Figure 6.4: Fulcrum drawing

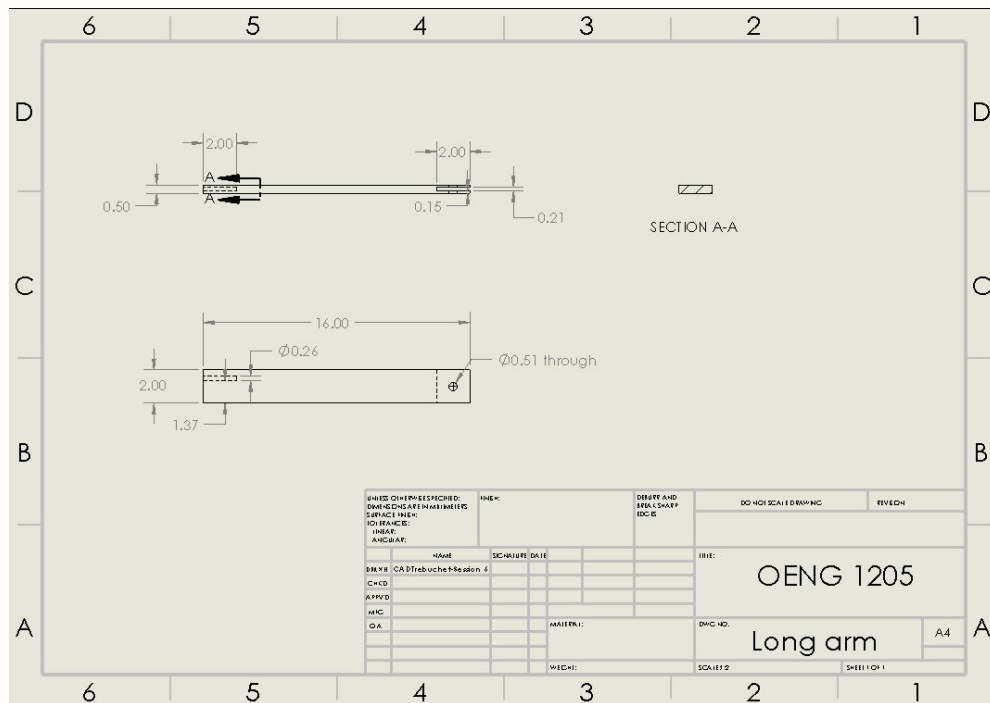


Figure 6.5: Long arm drawing

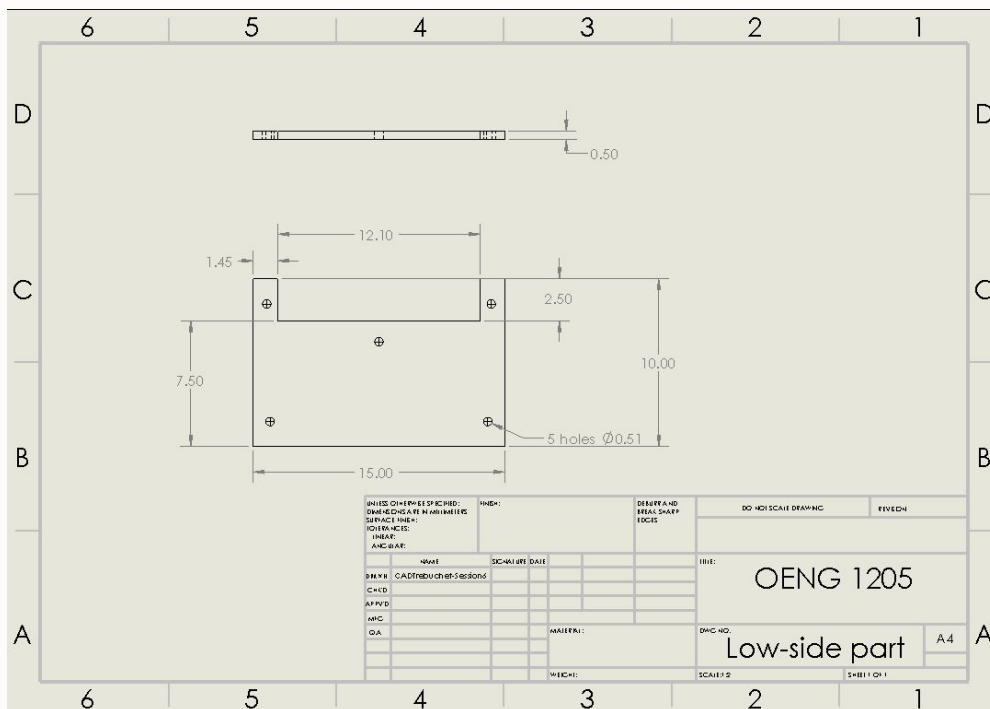


Figure 6.6: Low side drawing

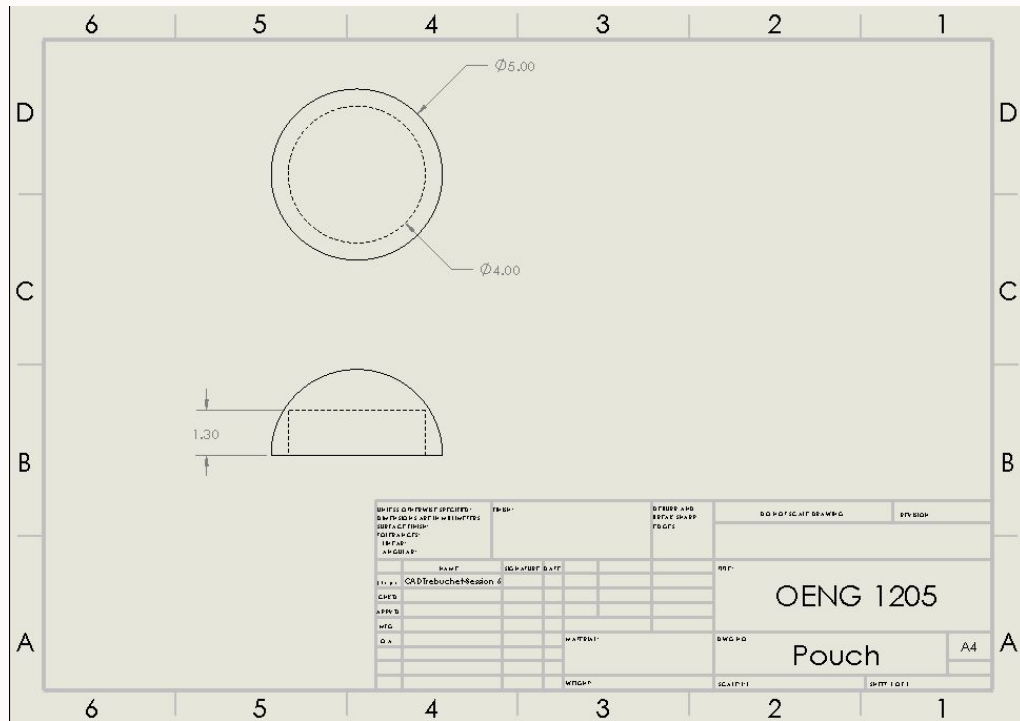


Figure 6.7: Pouch drawing

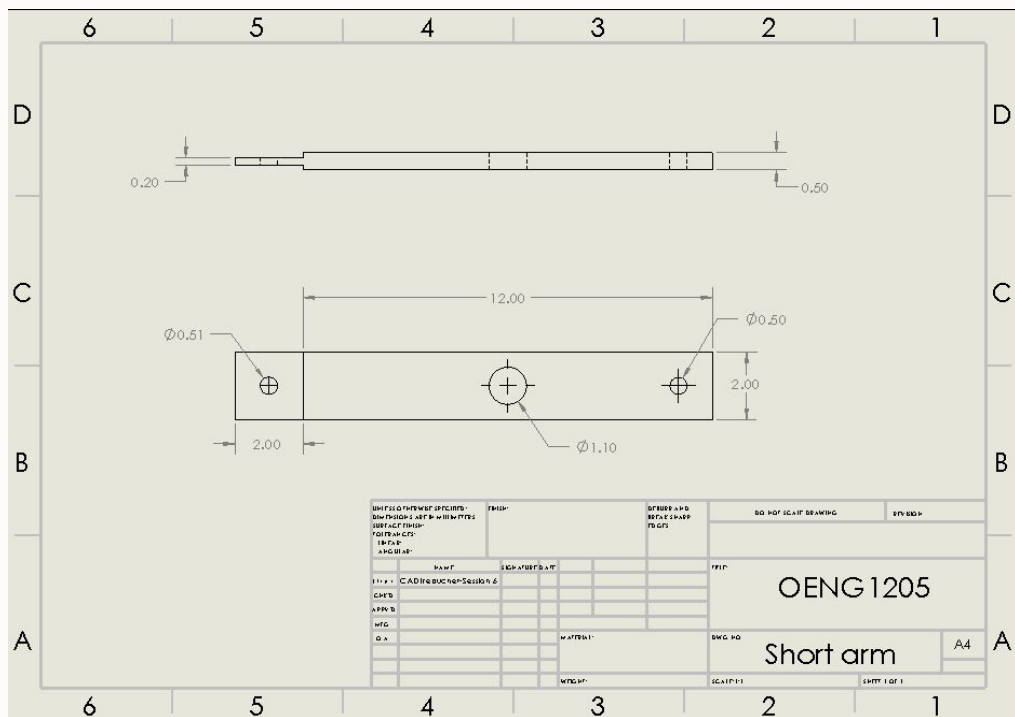


Figure 6.8: Short arm drawing

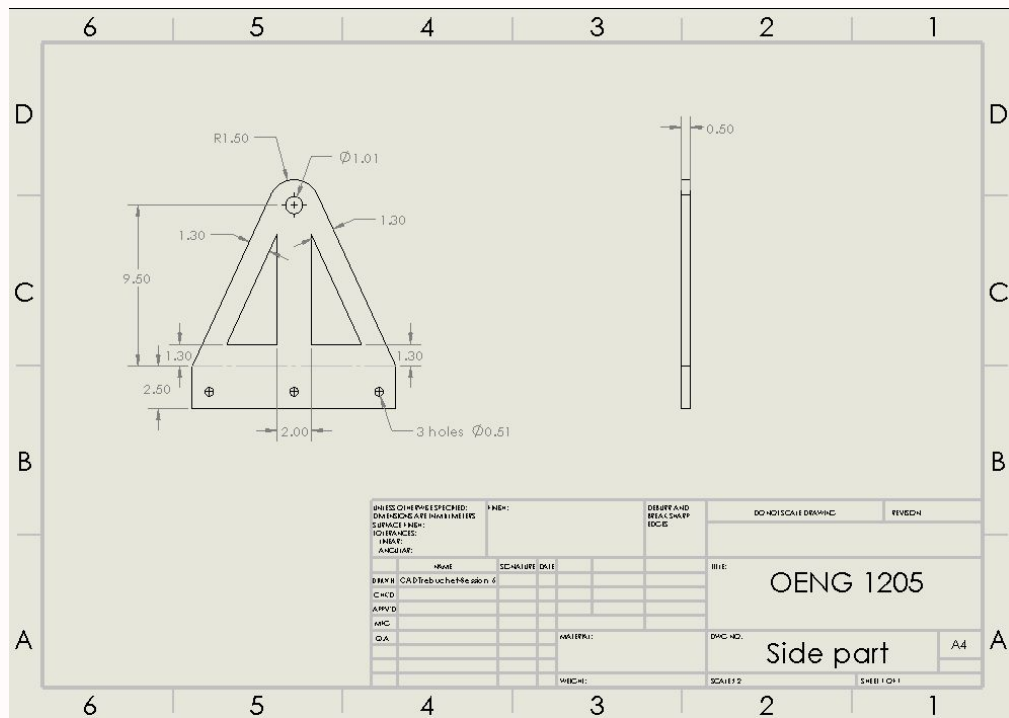


Figure 6.9: Side part drawing